Why Are Patients Waiting So Long to Get New Kidneys in Canada?

1. Introduction and Scope

1.1 Purpose

This project will develop an interactive Power BI dashboard to examine kidney transplant wait-list trends in Canada and compare selected indicators with the United States. The system will combine data from the Canadian Institute for Health Information (CIHI), the U.S. Organ Procurement and Transplantation Network/Scientific Registry of Transplant Recipients (OPTN/SRTR) and selected provincial organ donation policy sources. The goal is to help users identify trends, regional differences, and possible barriers to timely transplantation.

1.2 Objectives

The system will track kidney transplant and wait-list trends, compare provincial and regional indicators, examine donor trends and transplant outcomes, and benchmark Canadian data against suitable U.S. indicators. The findings will be presented through interactive visualizations to support evidence-based planning and public awareness.

1.3 Project Scope

Included

- Collecting data from approved public sources.
- Cleaning and standardizing available datasets using Python.
- Storing structured data in a MySQL database.
- Analyzing trends and preparing comparable indicators.
- Developing an interactive Power BI dashboard.
- Validating calculations, filters, and visualizations.
- Documenting data sources, limitations, and findings.

Excluded

- Providing medical advice or treatment recommendations.
- Predicting an individual patient's transplant date or survival.
- Allocating organs or making clinical decisions.
- Accessing private patient records.
- Claiming that observed associations prove the causes of transplant delays.
- Guaranteeing that every Canadian province and U.S. state can be compared for every indicator.

The project will use public and aggregate data. Comparisons will be limited to indicators with suitable definitions, coverage, and reporting periods.

2. Stakeholder Roles

Stakeholder Group	Needs and Responsibilities
Patients & Families	Access clear, aggregate information about wait-list and transplant trends.
Healthcare Administrators & Policy Planners	Review trends, identify areas for further investigation, and support resource planning.
Marketing & Public Relations Teams	Use regional indicators to guide awareness campaigns.
Organ Donation Organizations	Examine donor and transplant trends to inform outreach and planning.
Data analyst and project manager	Define indicators, analyze results, coordinate tasks, and document findings.
Data engineer	Collect, clean, standardize, and integrate data; maintain database quality.
Dashboard developer	Build dashboard pages, filters, visualizations, and supporting documentation.

The dashboard is intended for research and planning. It will not replace professional medical advice or official transplant guidance.

3. Functional Requirements

Functional requirements define what the system must do.

Priority levels:
Must = *Essential for delivery*;
Should = *Important if feasible;*
Could = *Optional enhancement;*

ID	Requirement	Priority
FR-01	The system shall programmatically collect and import kidney transplant, wait-list, donor, and outcome datasets from approved public sources, including structured national files (CIHI Excel/CSV), semi-structured international registries (US OPTN/SRTR JSON/APIs), and unstructured provincial policy text via web scraping.	Must
FR-02	The system shall record each dataset's source, reporting period, and retrieval date.	Must
FR-03	The system shall identify missing values, duplicate records, and inconsistent formats during data preparation.	Must
FR-04	The system shall standardize relevant field names, date formats, geographic labels, and categories.	Must
FR-05	The system shall store validated, structured data in a MySQL database.	Must
FR-06	The system shall support queries for the indicators selected for the dashboard.	Must
FR-07	The system shall display available kidney transplant and wait-list trends over time.	Must
FR-08	The system shall allow users to filter dashboard results by reporting year and Canadian province or region, where available.	Must
FR-09	The system shall display available donors and transplant indicators.	Must
FR-10	The system shall present Canada-U.S. comparisons only when the selected indicators have sufficiently compatible definitions and reporting periods.	Must
FR-11	The system shall display data sources, reporting periods, indicator definitions, and known limitations.	Must
FR-12	The system shall allow users to reset dashboard filters to their default state.	Should
FR-13	The system should allow users to export supported dashboard data or reports for further review.	Should
FR-14	The system could include additional policy context from provincial organ donation websites where reliable information is available.	Could

4. Non-Functional Requirements

Non-functional requirements define how well the system must operate.

ID	Requirement	Priority
NFR-01: Data quality	The system shall validate required fields, data types, duplicates, and key relationships before data is approved for analysis.	Must
NFR-02: Accuracy	Dashboard totals and calculated indicators shall match independently verified calculations based on the same source data and definitions.	Must
NFR-03: Performance	On the agreed test environment, standard dashboard pages shall load within 10 seconds under normal conditions, excluding initial data refresh and network delays.	Should
NFR-04: Usability	Dashboard pages shall use clear titles, labels, legends, and units. Users shall be able to identify the selected filters and reporting period.	Must
NFR-05: Security	The project shall use public, aggregate data, and shall not intentionally collect or display identifiable patient information.	Must
NFR-06: Data integrity	The database shall use appropriate primary keys, data types, and integrity checks to reduce duplication and invalid relationships.	Must
NFR-07: Traceability	Each reported indicator shall be traceable to its source dataset, reporting period, and documented calculation.	Must
NFR-08: Compatibility	The dashboard shall be tested in the supported Power BI environment agreed upon by the project team.	Must
NFR-09: Maintainability	Python scripts, database definitions, and project documentation shall be stored in the team's version-controlled GitHub repository.	Must
NFR-10: Compliance	The team shall follow applicable data-source terms, attribution requirements, and relevant institutional rules.	Must

Performance targets will be tested in the agreed project environment. They are project targets, not claims about current system performance.

5. Acceptance Criteria

A requirement is accepted when its corresponding test passes.

ID	Acceptance criteria
AC-01	All selected datasets are imported successfully, and their source and reporting period are documented.
AC-02	Data-quality checks produce a record of missing values, duplicates, and format issues.
AC-03	Standardized fields follow the documented data definitions and formats.
AC-04	Validated records load into MySQL, and database checks identify invalid key relationships.
AC-05	Selected dashboard indicators match independently calculated results for the same dataset and reporting period.
AC-06	Users can filter available results by year and province or region, and all displayed metrics update correctly.
AC-07	Donor, transplant, and wait-list visuals display the selected indicators, labels, units, and reporting periods accurately without missing values..
AC-08	Every Canada-U.S. comparison included in the dashboard has documented definitions and reporting periods. Incompatible indicators are excluded or clearly flagged.
AC-09	Dashboard pages identify their data sources and disclose relevant gaps or limitations.
AC-10	The dashboard's reset control restores the default filters.
AC-11	If export is implemented, a test export opens successfully and contains the selected data or report.
AC-12	A performance test confirms that standard dashboard pages meet the 10-second target in the agreed test environment, if that target is retained.
AC-13	The repository contains the project scripts, database definitions, and documentation required to reproduce the workflow.
AC-14	Final review confirms that no identifiable patient information is intentionally displayed, and that required source of attribution is present.

The project team shall record each test as **Pass**, **Fail**, or **Not Applicable**, with a brief note on any issue found. Optional requirements that are not implemented shall be recorded as Not Applicable or deferred, as appropriate.

6. Data and Reporting Constraints

The system depends on the coverage and quality of publicly available data.

- **Availability:** Some indicators may not be reported consistently across provinces, states, or years.
- **Comparability:** Differences in definitions and reporting methods may limit cross-border comparisons.

- **Missing data:** Missing values shall be documented and shall not be treated as zero unless the source definition supports that interpretation.
- **Interpretation:** The dashboard may reveal patterns associated with longer waits, but it cannot establish causation on its own.
- **Policy context:** Web-sourced policy information shall be linked to its source and review date where included.

These constraints will guide indicator selection and the wording of the final findings.

7. Revision History and Approval

Version	Date	Change	Status
1.0	Sep 27, 2026	Initial project proposal submitted.	Returned for Revision
1.1	October 2, 2026	Revised proposal to address data integration, web scraping, and MySQL implementation.	Resubmitted
1.1	October 4, 2026	Revised proposal reviewed; team authorized to proceed.	Approved to proceed
1.0	October 23, 2026	User Requirements and Analysis Document submitted.	Awaiting feedback

Changes to scope, requirements, priorities, or acceptance criteria shall be recorded in this table. The team shall update the version number when a revised document is submitted for review.

8. Definition of Project Success

The project will be considered successful when:

1. The selected public datasets are collected, documented, cleaned, and stored.
2. The dashboard presents accurate, interactive views of the available kidney transplant indicators.
3. Supported Canada-U.S. comparisons use documented and compatible measures.
4. The acceptance tests for all Must requirements pass.
5. The final dashboard and supporting documentation disclose relevant data limitations.

The final deliverable will provide a clear view of kidney transplant trends and evidence to support further research and planning. It will not claim to determine a single cause of Canada's transplant delays or predict individual patient outcomes.

References

Canadian Institute for Health Information. (2025). *Median time from dialysis start to kidney transplantation* [Indicator]. https://www.cihi.ca/en/indicators/median-time-from-dialysis-start-to-kidney-transplantation

Canadian Institute for Health Information. (2025). *Summary statistics on organ transplants, wait-lists and donors* [Data tables]. https://www.cihi.ca/en/summary-statistics-on-organ-transplants-wait-lists-and-donors

Canadian Institute for Health Information. (2025). *Organ donation and transplantation infographics*. https://www.cihi.ca/en/organ-donation-and-transplantation-infographics

Health Resources and Services Administration. (2025). *Organ Procurement and Transplantation Network (OPTN) data building portal*. U.S. Department of Health and Human Services. https://www.hrsa.gov/net/optn?base=hrsa&path=/data/view-data-reports/build-advanced

Scientific Registry of Transplant Recipients. (2024). *OPTN/SRTR annual data report*. Health Resources and Services Administration. https://srtr.hrsa.gov/transplant-professionals/optnsrtr-annual-data-report/